Who takes which department, and how they do: a comparison across batches

Summary.
The question we tested: does how women perform, compared with men, in a subject in the first four semesters line up with how many women later take that department?
Findings:

1. **Across subjects, it does.** The two departments with the fewest women, physics and mathematics, are the only two where men score higher; biology, with the most women, has the largest lead for women (correlation 0.73, rank correlation 0.90 across five departments). With only five departments this is not statistically conclusive (exact p = 0.08), but the direction is clear and no single department drives it.
2. **Within a department across batches, the relationship runs the other way:** in batches where a department had more women, women's lead over men was smaller (about −0.16 grade points per 10 extra points of women's share, p ≈ 0.04), the same pattern as in whole batches (correlation 0.86 at semester 4, p = 0.007). Meanwhile, physics has lost about half its share of women while biology has grown to about half. These are real gaps, and the data cannot say why.

1. The data

The data set

We next received the male-to-female ratio (men per woman) in each of five departments for eight batches, with an overall ratio for each batch, from our DoAA. The table gives ratios only, not counts, so we cannot say how many students are behind each number.

Men per woman	Batch 1	2	3	4	5	6	7	8	Average
Biology	1.89	1.00	2.00	1.67	1.37	1.50	0.85	0.79	1.38
Chemistry	4.50	2.91	2.71	4.40	1.44	1.70	1.94	2.42	2.75
Earth sciences	3.29	3.44	2.83	8.25	2.80	12.00	1.40	8.00	5.25
Mathematics	5.50	8.00	12.00	4.00	4.25	5.50	–	12.00	7.32
Physics	2.20	3.46	5.38	5.78	5.75	18.33	5.40	6.22	6.56
Whole batch	2.85	2.58	3.42	4.05	2.42	3.47	2.08	2.70	2.95

Women's share of each department

A ratio of men per woman is hard to read, so we convert each one to the share of women: 100 ÷ (1 + ratio). A ratio of 1.00 is 50% women; 3.00 is 25%; 12.00 is about 8%.

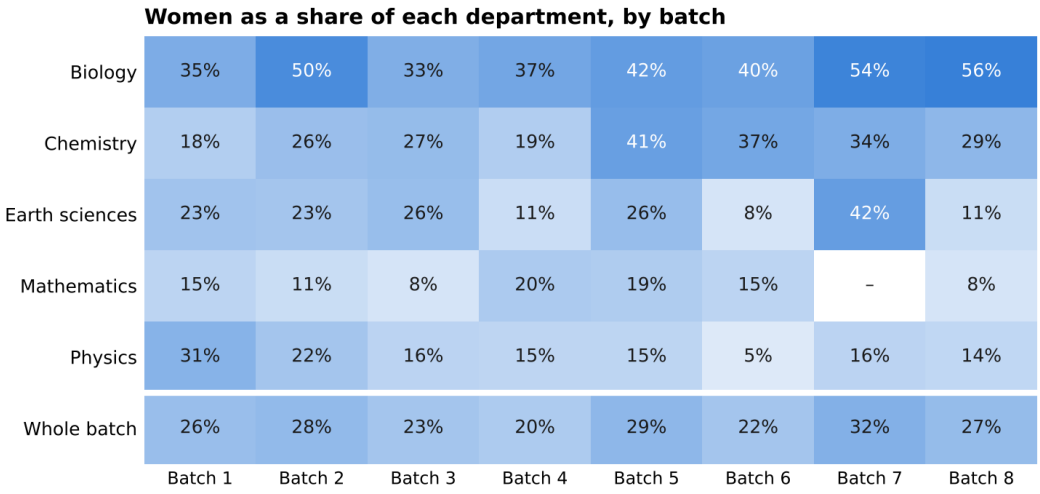


Figure 1. Women as a share of each department, by batch. Darker blue means more women. The last row is the whole batch.

The whole batch is between 20% and 33% women, and is about 25% on average. That is close to the 27% women among IISER Kolkata undergraduates in our earlier report, which was counted differently and in a different year. The departments differ far more than the batches do: biology is 33% to 56% women, chemistry 18% to 41%, earth sciences 8% to 42%, physics 5% to 31% and mathematics 8% to 20%.

2. How representation has moved across batches

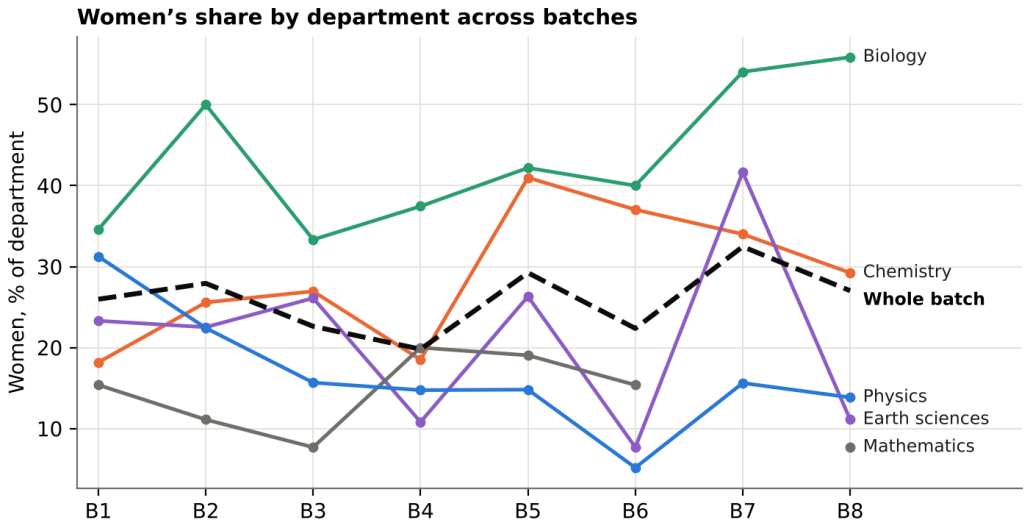


Figure 2. Women's share of each department across batches (B1 oldest, B8 newest). The dashed line is the whole batch.

Department	Average of first three batches	Average of last three batches	Trend (points per batch)	How firm
Biology	39%	50%	+2.3	Fairly consistent (rank correlation 0.69, p = 0.06)
Chemistry	24%	33%	+2.1	Rose, then fell after Batch 5 (0.69, p = 0.06)
Earth sciences	24%	20%	−0.4	Very uneven; no trend (−0.05)
Mathematics	11%	14%	−0.3	No trend; small numbers, one missing batch
Physics	23%	12%	−2.2	Clear fall (−0.79, p = 0.02)
Whole batch	26% (first three)	27% (last three)	+0.5	No trend (0.24, p = 0.57)

Two things stand out. Physics, which already had few women, has lost about half its share, from roughly a quarter to about one in nine. Biology has grown the other way and is now the department where women are most likely to be the majority. The whole-batch average hides both. If departments are chosen by the students, it matters that the movement is in opposite directions: it suggests that something in how students choose, or are guided toward, departments is changing, and that is worth contemplating. We have not looked into why.

Caution: percentages are computed from ratios, and a small department can swing by many points when a few students move. The trends are to be treated as a direction, not an estimate.

3. How men and women did: the last subject GPA data

The earlier "Subject GPA by gender" report gives the mean GPA of men and women in each of seven subject areas, in the first four semesters, for six batches (Batches 3 to 8). These are grades in courses that every student takes before choosing a major, so they describe all students, not only those who later join the department. Orange below means women scored higher; blue means men did.

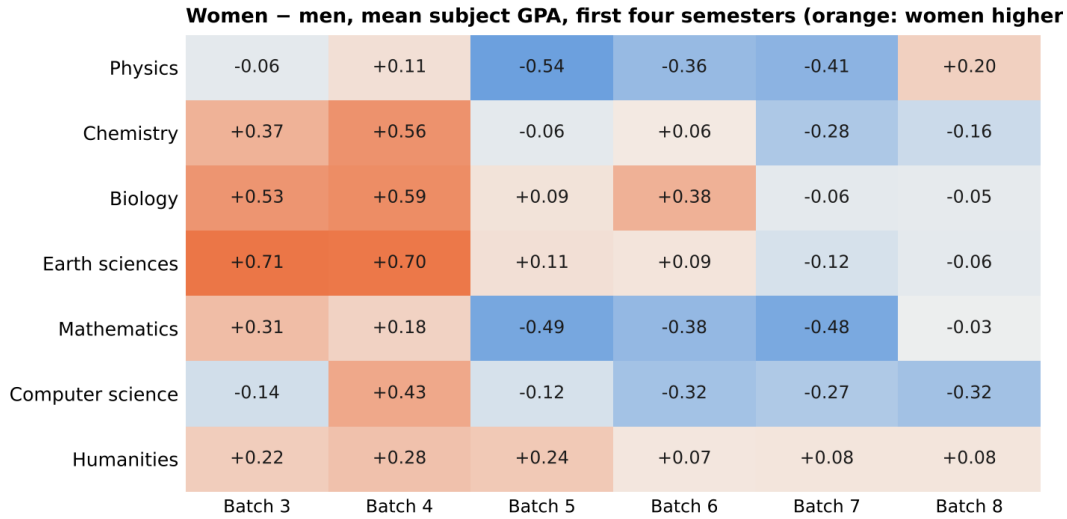


Figure 3. Women minus men in mean subject GPA, by subject and batch (first four semesters).

Subject	Women − men (average of 6 batches)	Batches in which women scored higher
Humanities	+0.16	6 of 6
Biology	+0.25	4 of 6
Earth sciences	+0.24	4 of 6
Chemistry	+0.08	3 of 6
Mathematics	−0.15	2 of 6
Computer science	−0.12	1 of 6
Physics	−0.18	2 of 6

The gaps are small, mostly under a fifth of a grade point on average, and they change sign from batch to batch. Humanities is the only subject where women were ahead in every batch.

4. The question: do grades in the first four semesters line up with who takes the department?

Five subjects have a department ratio: physics, chemistry, biology, earth sciences and mathematics. Computer science and humanities have none and are left out here. We looked at the question in three ways, because each answers something different. We use women's share of the department (rather than men per woman) wherever we fit lines, since the ratio is very skewed (from 0.8 to 18) and a few extreme batches would dominate. Where the choice changes the result, we mention the same.

A. Across subjects: do the subjects where women do worse have fewer women?

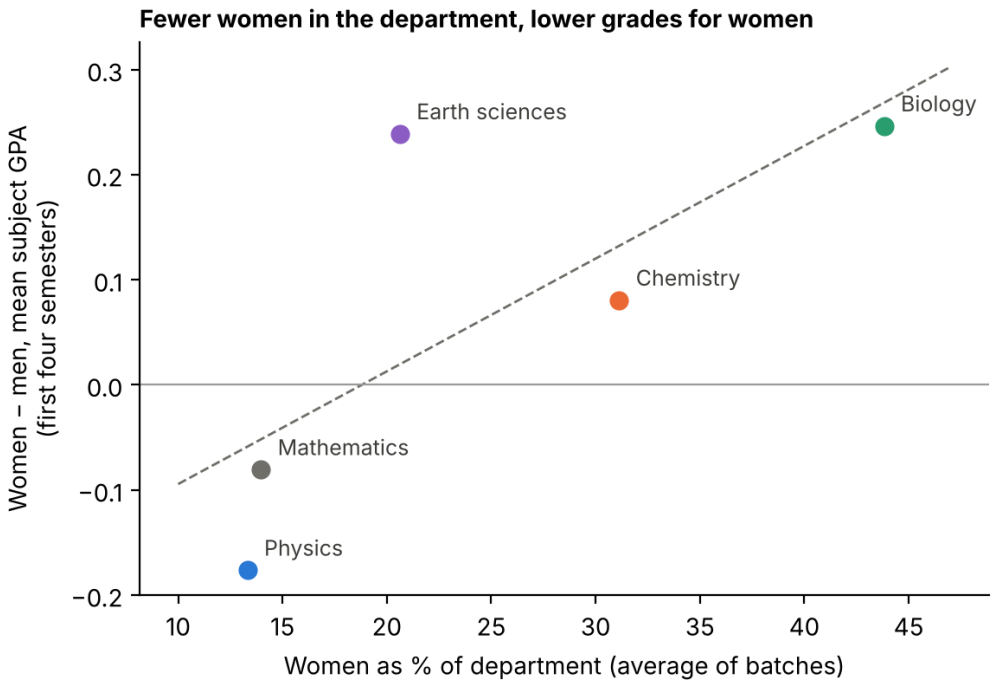


Figure 4. Each dot is one department: its average share of women across batches against the average gap in women's and men's subject GPA in the first four semesters.

Department	Women's average share	Women – men GPA gap	Batches where women ahead
Biology	44%	+0.25	4 of 6
Earth sciences	21%	+0.24	4 of 6
Chemistry	31%	+0.08	3 of 6
Mathematics	14%	−0.08 (5 batches)	2 of 6 (−0.15 over six)
Physics	13%	−0.18	2 of 6

The two departments where men score higher, physics and mathematics, are the two with the fewest women, and the department with the most women, biology, has the biggest lead for women. Fitting a line, each 10 points more women goes with women's grades being about 0.11 grade points higher relative to men (Pearson 0.73, rank correlation 0.90). With five points the exact p-value for the rank correlation is 0.08, and the smallest it could be with five points is 0.017, so this test could never be strongly conclusive. The pattern holds if any one department is dropped (rank correlation 0.8 to 1.0). Earth sciences is the clear exception: few women (21%), yet women score higher.

B. Within a department, across batches: do batches with more women look different?

This tests the same idea over time. For each department we ask whether, in batches where its share of women was higher than usual, women's relative grades were higher or lower than usual. We pool the five departments (29 department-by-batch points) and compare each point with its own department's average, so differences between subjects drop out.

Measure of take-up	Result	Reading
Women's share of the department	−0.16 GPA points per +10 points of share (p = 0.03 to 0.04)	In batches where more women took the department, women's lead over men was smaller
Men per woman (the raw ratio)	+0.01 per extra man per woman (p = 0.6)	No relationship, but the ratio is highly skewed, so the share is the better measure

This result is sensitive to how take-up is measured, so we treat it as a lead and not a firm finding. It goes the opposite way to part A. One reading is that the two are different effects: part A is about the subject (its culture, how it is taught, who feels at home in it), and part B is about selection (when fewer women join a department, the women who do may be stronger). Both readings are hypotheses; this data cannot separate them.

C. Whole batches

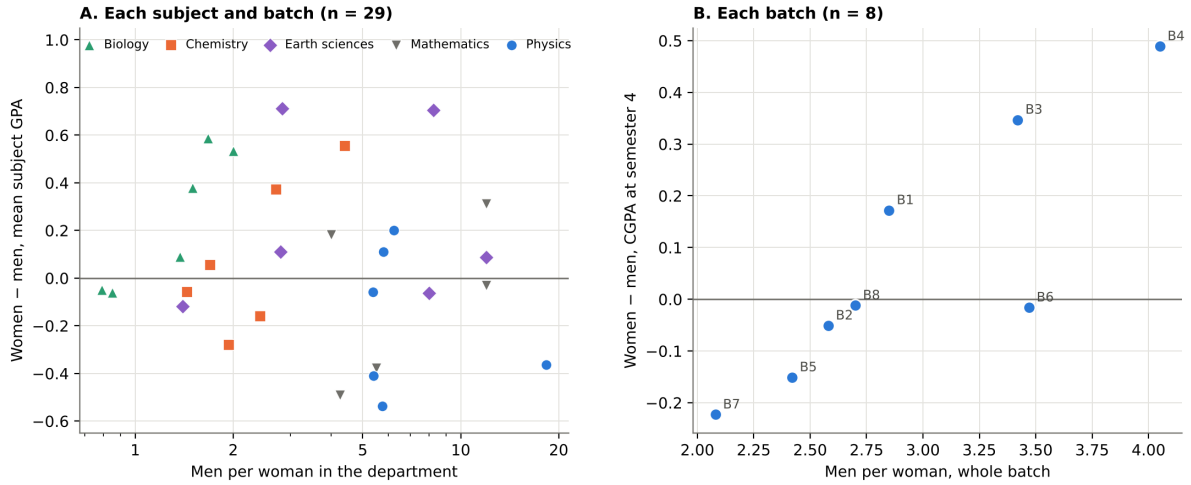


Figure 5. A: each dot is one subject in one batch; the horizontal axis is logarithmic. B: each dot is one batch (B1 oldest). Above the line, women's mean grade is higher.

Using the CGPA gap from the "Gender and CGPA across batches" report, **batches with more men per woman have a larger lead for women**: correlation 0.86 at semester 4 (p = 0.007, n = 8), 0.71 at semester 10 (p = 0.047) and 0.76 across semesters 1 to 4 (p = 0.03). Dropping any one batch leaves the semester 4 result at 0.79 or higher. Batch 4 (about four men per woman) has the largest lead for women (+0.49), and Batch 7 (about two) has men ahead (−0.22). This is the same direction as part B.

All the tests together

Test	n	Result	Reading
Across the five departments (rank correlation)	5	0.90 (exact p = 0.08)	Fewer women, lower relative grades for women; direction clear, but not conclusive
Across the five departments (Pearson)	5	0.73 (p = 0.16)	The same
Within department, share of women	29	−0.16 per 10 points (p = 0.03–0.04)	More women, smaller lead for women
Within department, men per woman	29	p = 0.6	No relationship on this scale
All 29 points pooled, ratio against gap (rank)	29	0.06 (p = 0.75)	Mixes the two opposite effects above, so it cancels out
Whole batch, men per woman, semester 4	8	0.86 (p = 0.007)	More men, women do relatively better
Whole batch, semester 10	8	0.71 (p = 0.047)	The same, weaker
Women's share against women's own grades, pooled	29	0.42 (p = 0.02)	Largely the subject effect: fades to −0.20 within a department

What the pooled 29-point result (0.06) hides: the across-subject and within-subject effects have opposite signs, so pooling them gives roughly zero.

How much weight to give these results: we ran about nine tests on small samples, so some of the p-values below 0.05 could be chance, and none would survive a strict correction for multiple testing. The

across-subject pattern and the whole-batch pattern are the two that need follow-up, and the first is limited by having only five departments.

5. Comparing with our two earlier reports

Earlier finding	What the new data show	Agree?
Gender and CGPA across batches: women lead at semester 4 in 3 of 8 batches (+0.07 on average) and at semester 10 in 6 of 8 (+0.14).	The same CGPA data. The size of the gap follows the batch's ratio of men to women (Section 4C): women lead in batches with more men.	Yes, and it adds a pattern
Subject GPA by gender: women ahead in humanities, biology and earth sciences; behind in physics, mathematics and computer science.	Same order in the six-batch averages. Chemistry is close to equal. The subjects where women trail are the departments with the fewest women (Section 4A).	Yes, and it links the two
Women are about 27% of IISER Kolkata undergraduates, the lowest of the IISERs; 1 in 30 physics faculty is a woman.	The whole batch is about 25% women. Physics is where women's share has fallen most, and where faculty are least likely to be women.	Yes

What would make the analysis stronger

- **Counts,** of men and women in each department and batch, to see how big the groups are.
- **Grades by department after the major** (semesters 5 to 10), so grades and ratios describe the same students. This would be the single most useful data point.
- **Admission ranks or entrance scores** by gender, to test the selection hypothesis.
- **Interviews and a short survey** on why students chose or avoided physics, mathematics and computer science.